Sheet 47 Mon. 8 March 2027 Reciprocal function: variations; relative position of y = x and y = x^2

1. Warm-up · Flash [CAL] Calculate $\frac{1}{2}$, $\frac{1}{-4}$ and $\frac{1}{0{,}1}$.

Ans. 0,5 ; −0,25 ; 10.

2. Core · Method [CMP] Compare $\frac{1}{3}$ & $\frac{1}{5}$; what can be said about the variations of the reciprocal function on $]0;+\infty[$?

Ans. $\frac{1}{3} > \frac{1}{5}$ while $3 < 5$: the reciprocal fn decreases on $]0;+\infty[$ (and on $]-\infty;0[$, but not on $\mathbb{R}^*$).

3. Core + · True/false [CMP] True or false? Justify. For every real number $x > 0$, $x^2 \geqslant x$.

Ans. False. Counterexample: $x = \frac{1}{2}$, $x^2 = \frac{1}{4} < \frac{1}{2}$. In fact $x^2 \geqslant x$ if and only if $x \geqslant 1$ (for $x > 0$).

4. Back · Flash [PER] What are the front view and the top view of a cylinder standing on its base?

Ans. Front view: a rectangle; top view: a disc.

5. Running thread · Prove [LIT] Show that the product of two even integers is a multiple of 4.

Ans. With $a = 2p$, $b = 2q$: $ab = 4pq$, multiple of 4.

Sheet 48 Tue. 9 March 2027 Compare f (a) and $f(b)$; optimisation problems

1. Warm-up · Flash [CMP] $f(x) = x^2$. Compare $f(-3)$ and $f(2)$.

Ans. $f(-3) = 9 > 4 = f(2)$.

2. Core · Method [REG] $f(x) = x^2$, $a = -1.5$ and $b = -2$. Compare $f(a)$ and $f(b)$ without calculation, using the variations.

Ans. $-2 < -1{,}5 \leqslant 0$ and f decreasing on $]-\infty;0]$: $f(-2) > f(-1{,}5)$. Check: 4 $> 2{,}25$.

3. Core + · Optimisation [MOD] $A(x) = x(10-x)$ for $x \in [0;10]$ (area of a rectangle with perimeter 20). Show that $A(x) = 25 - (x-5)^2$ and deduce the maximum.

Ans. $25-(x-5)^2 = 25 - x^2 + 10x - 25 = x(10-x)$; maximum 25 attained for $x = 5$ (square with side 5).

4. Back · Flash [STA] 18 out of 24 pupils have an average above 10. Give the relative frequency as a percentage.

Ans. $\frac{18}{24} = 75\,\%$.

5. Running thread · MCQ [EQ] The solution of $\frac{x}{3} + 1 = \frac{7}{2}$ is:

A. $\frac{15}{2}$ **B.** $\frac{9}{2}$ **C.** $\frac{21}{2}$ **D.** $\frac{5}{6}$

Ans. A. $\frac{15}{2}$

Pitfall: forgetting to subtract the 1 before multiplying by 3 ($\frac{21}{2}$).

Cross-tabulating two categorical variables

Sheet 49 Thu. 11 March 2027 Cross-tabulating two categorical variables

1. Warm-up · Flash [PRO] Express $\frac{24}{60}$ as a percentage.

Ans. 40 %.

2. Core · Flash [LEC] In a class of 30 pupils, 18 are girls. What is the frequency of girls? Of boys?

Ans. $\frac{18}{30} = 60\,\%$; 40 %.

3. Core + · Classify [STA] Classify each variable: eye colour; baccalauréat grade (AB, B, TB); height in cm.

Ans. Qualitative nominal; qualitative ordinal (natural order); quantitative.

4. Back · Problem [MOD] A rectangle with perimeter 20 cm has length x. Express its area $A(x)$; calculate $A(3)$ and $A(5)$.

Ans. Width $10 - x$; $A(x) = x(10-x)$; $A(3) = 21$, $A(5) = 25$ cm^2.

5. Running thread · Flash [SOL] Calculate the exact volume of a cylinder with radius 2 cm and height 5 cm.

Ans. $V = \pi \times 4 \times 5 = 20\pi$ cm^3.

Sheet 50 Mon. 15 March 2027 Qualitative variables; two-way frequency table

1. Warm-up · Flash [CAL] Complete: $12 + \ldots = 30$ and $22 + \ldots = 40$.

Ans. 18 ; 18.

2. Core · Table [LEC] Total 40 students. Girls: 22; half-boarders (DP): 25; DP girls: 15. Complete the two-way table.

	DP	Non-boarders	Total
Girls	15	…	22
Boys	…	…	…
Total	25	…	40

Ans. Non-boarder girls 7; DP boys 10; non-boarder boys 8; total boys 18; total non-boarders 15 (check: $7+8=15$).

3. Core + · True/false [LEC] True or false? Justify. "Satisfaction level (low, medium, high)" is a nominal qualitative variable.

Ans. False. There is a natural order: it is an ordinal qualitative variable.

4. Back · Problem [CMP] Discount A: -30% off 80,000 Ar. Discount B: -20% then -10%. Which is the better deal?

Ans. A: 56,000 Ar; B: $80{,}000 \times 0{,}8 \times 0{,}9 = 57{,}600$ Ar. A is cheaper.

5. Running thread · Method [DEM] We know that $AB^2 + AC^2 = BC^2$. What conclusion do we draw? What property?

Ans. ABC is right-angled at A : converse of Pythagoras' theorem.

Sheet 51 Tue. 16 March 2027 Marginal frequencies

1. Warm-up · Flash [PRO] Express $\frac{25}{40}$ as a percentage.

Ans. 62,5 %.

2. Core · Method [LEC] With the table from sheet 50: calculate the marginal frequency of girls and that of the half-boarders.

Ans. Girls: $\frac{22}{\text{—}} = 55\%$; DP: $25\text{—} = 62.5\%$ (marginal count over the total count).

Marginal frequency = row (or column) total / overall total.

3. Core + · Flash [LEC] The marginal frequency of girls is 55%. What is that of boys?

Ans. 45%: the marginal frequencies of one variable add up to 100%.

4. Back · Flash [SUB] Calculate $(2a-b)^2$ for $a = 3$ and $b = -2$.

Ans. $(6+2)^2 = 64$.

5. Running thread · Flash [REP] $A(-1;2)$ and $B(3;2)$. What can be said about the line (AB) ? Give AB.

Ans. Horizontal (same y-coordinate), parallel to the x-axis; $AB = 3-(-1) = 4$.

Sheet 52 Thu. 18 March 2027 Conditional frequencies

1. Warm-up · Flash [PRO] Give $\frac{15}{22}$ as a percentage rounded to one decimal place.

Ans. 68,2 %.

2. Core · Method [LEC] With the same table: among the girls, what is the frequency of DP pupils? Among the boys?

Ans. Girls: $\frac{15}{22} \approx 68.2\%$; boys: $10 \quad \frac{}{18} \approx 55.6\%$ (divide by the total of the subpopulation).

3. Core + · Distinguish [LEC] Distinguish "the frequency of girls among DP pupils" from "the frequency of DP pupils among the girls".

Ans. $\frac{15}{25} = 60\%$ and $\frac{15}{22} \approx 68.2\%$: the reference total is not the same.

4. Back · Construction steps [CON] We draw $[AB]$ of 6 cm, then the circles with centres A and B and radius 4 cm, which intersect at C and D. Type of $ACBD$?

Ans. A rhombus: $AC = CB = BD = DA = 4$ cm (four equal sides).

5. Running thread · Flash [DIV] Is 91 prime?

Ans. No: $91 = 7 \times 13$.

Pitfall: it is divisible neither by 2, nor by 3, nor by 5.

Sheet 53 Mon. 22 March 2027 Complete a two-way table

1. Warm-up · Flash [PRO] Calculate 35% of 80.

Ans. 28.

2. Core · Method [LEC] 200 customers: 60% women; 25% of the women and 50% of the men buy online. Complete the two-way table.

Ans. Women 120, of whom 30 online; men 80, of whom 40 online; online in total: 70.

We move from conditional frequencies to counts by multiplying by the size of the subpopulation.

3. Core + · Check [LEC] A table states "row total: $45 + 55 = 100$" but "column total: $60 + 30 = 90$". What can be concluded?

Ans. The table is inconsistent: the grand total must be the same by rows and by columns; there is a data-entry or calculation error.

4. Back · True/false [PRB] True or false? Justify. After three "heads" in a row, the probability of getting "tails" on the next toss is greater than $\frac{1}{2}$.

Ans. False. The tosses are independent: the probability remains $\frac{1}{2}$.

5. Running thread · Flash [GEO] ABC right-angled at A, $BC = 10$ cm, $\widehat{ABC} = 30°$. Calculate AC.

Ans. $AC = BC \times \sin 30° = 10 \times 0{,}5 = 5$ cm.

Use sine: opposite side / hypotenuse.

Sheet 54 Tue. 23 March 2027 Real data: processing a file and filtering (AND, OR, NOT)

1. Warm-up · Flash [LEC] In a file of 1,000 rows, 400 satisfy A, 300 satisfy B and 100 satisfy A and B. How many satisfy "A AND B" ? "A OR B" ?

Ans. 100 ; $400 + 300 - 100 = 600$ (the intersection is not counted twice).

2. Core · Method [LEC] With the same data: how many satisfy "NOT A" ? "A AND NOT B" ?

Ans. $1\,000 - 400 = 600$; $400 - 100 = 300$.

3. Core + · Reading code [ALG] What is the final value of compte ?

```
compte = 0
for age in [15, 16, 17, 16, 18]:
    if age >= 16 and age < 18:
        compte = compte + 1
```

Ans. 3 (ages 16, 17, 16).

4. Back · Flash [NUM] Give the decimal expansion of $\frac{7}{8}$ then the irreducible fraction equal to 0.12.

Ans. 0,875 ; $\frac{12}{100} = \frac{3}{25}$.

5. Running thread · Formula → table [REG] $f(x) = x^2 - 2x$. Complete a table for $x \in \{0 ; 1 ; 2 ; 3\}$.

Ans. 0 ; −1 ; 0 ; 3.